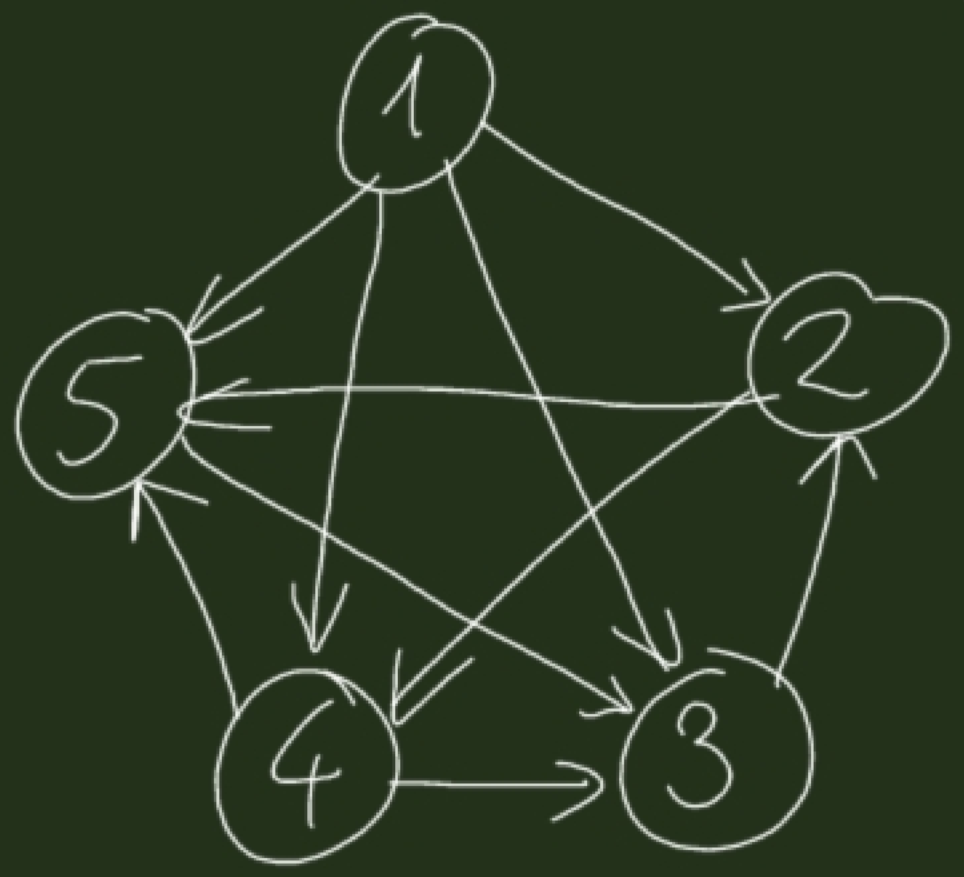
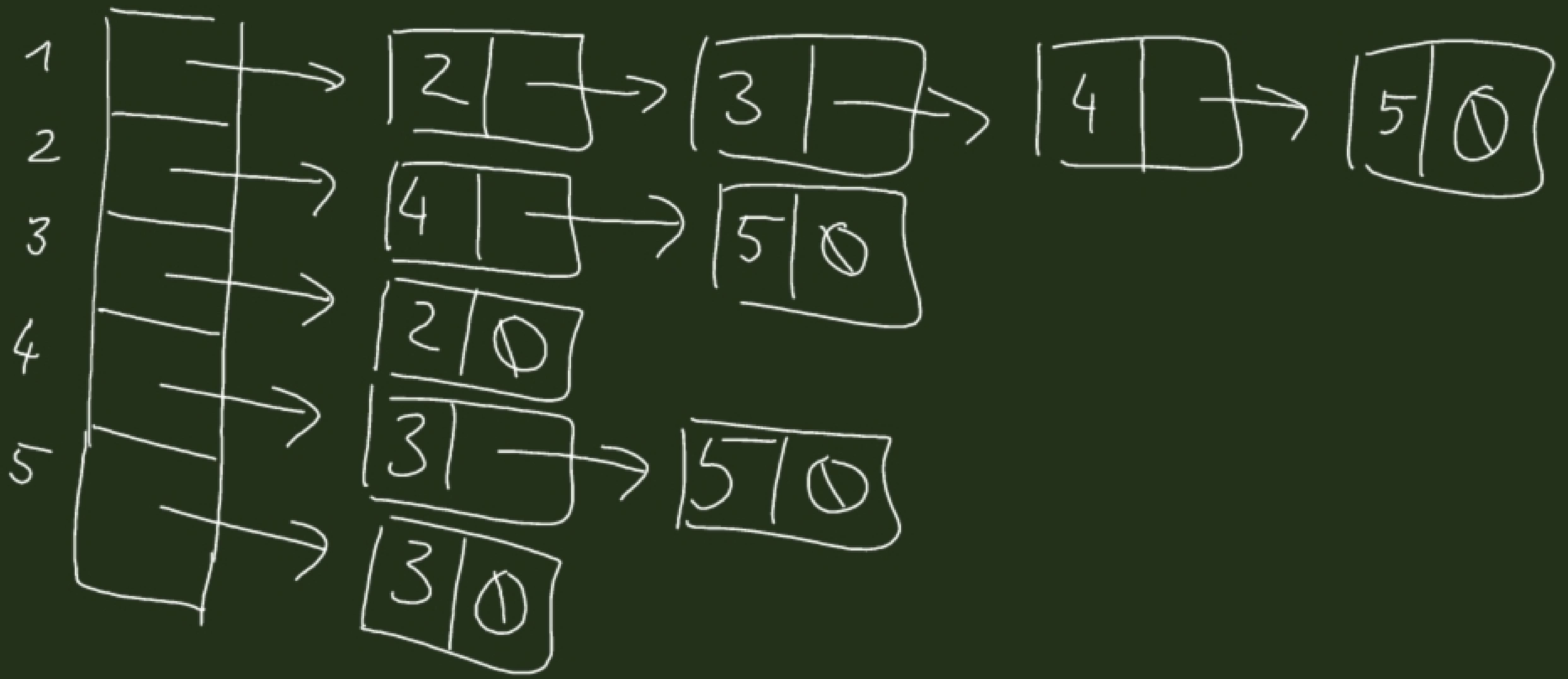


Adj. matrix:

	1	2	3	4	5
1	0	1	1	1	1
2	0	0	0	1	1
3	0	1	0	0	0
4	0	0	1	0	1
5	0	0	1	0	0

Textual:

1 → 2; 3; 4; 5.
2 → 4; 5.
3 → 2.
4 → 3; 5.
5 → 3.



S1L

Adj. list:

1 $\rightarrow$ 2 ; 4 ; 7 ; 8.

2 $\rightarrow$ 3 ; 7.

3 $\rightarrow$ 1 ; 5.

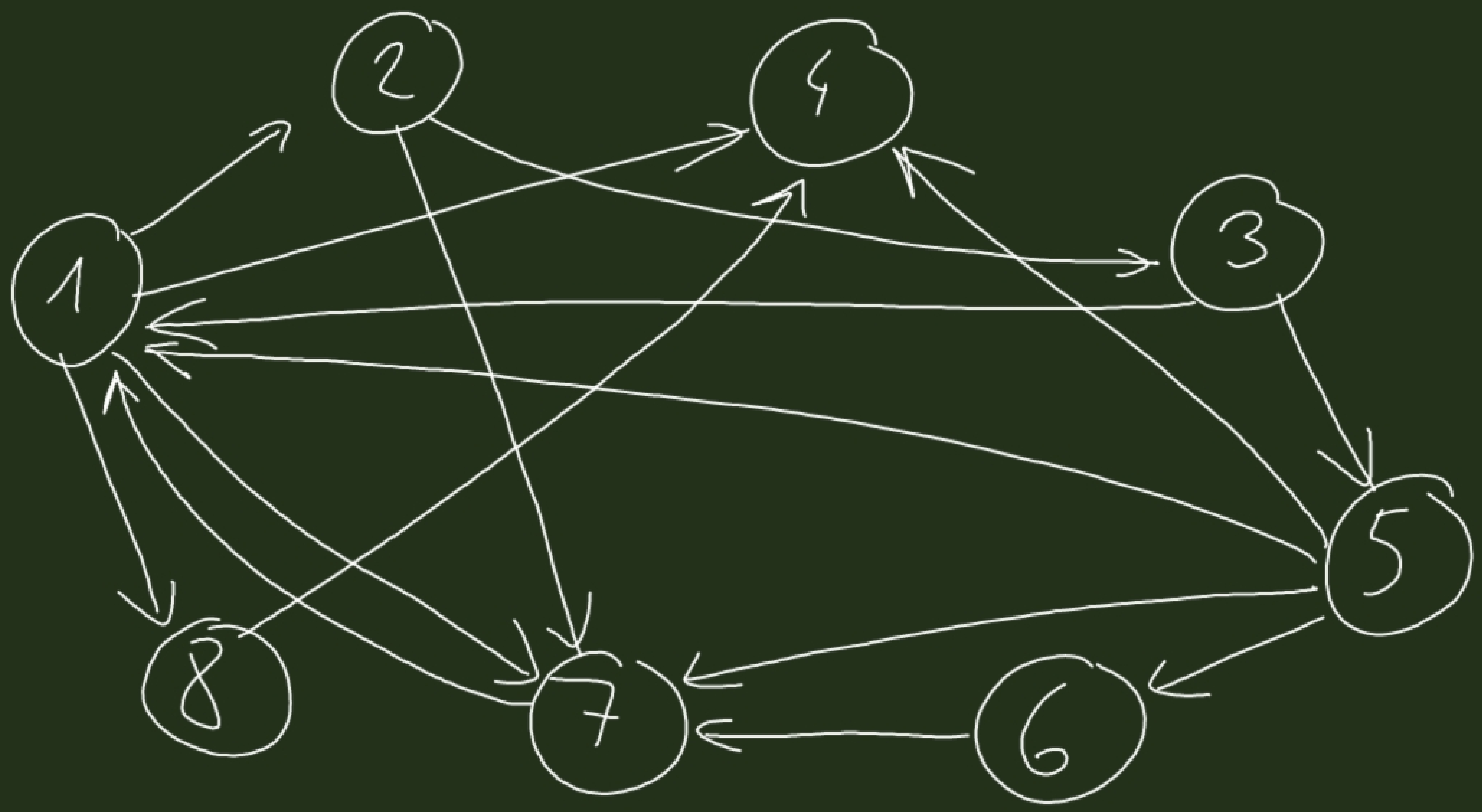
4 $\rightarrow$.

5 $\rightarrow$ 1 ; 4 ; 6 ; 7.

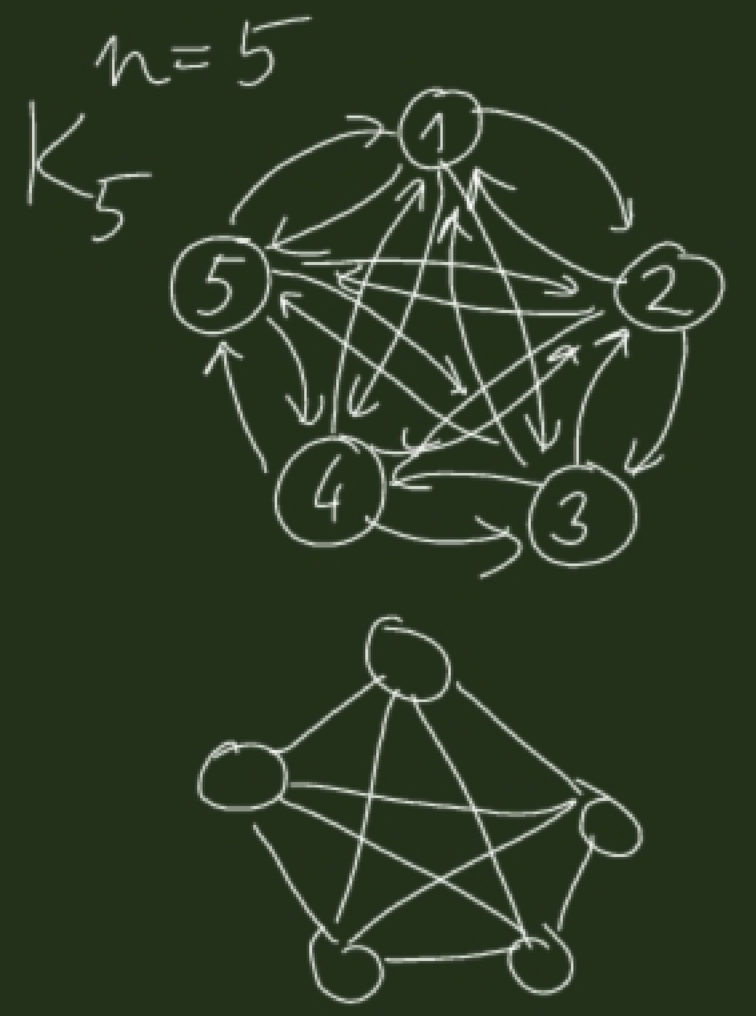
6 $\rightarrow$ 7.

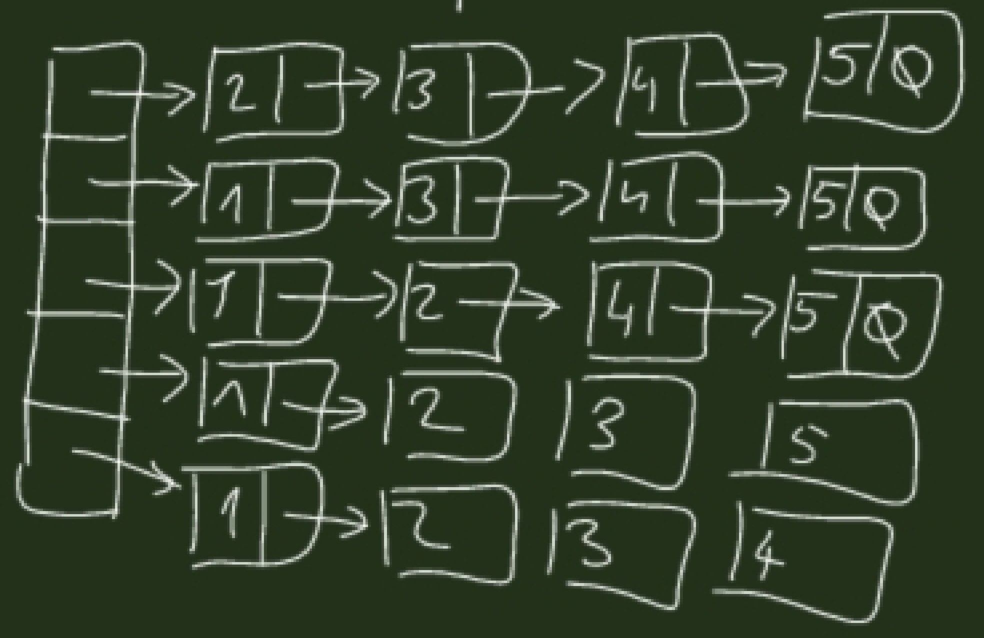
7 $\rightarrow$ 1.

8 $\rightarrow$ 4.



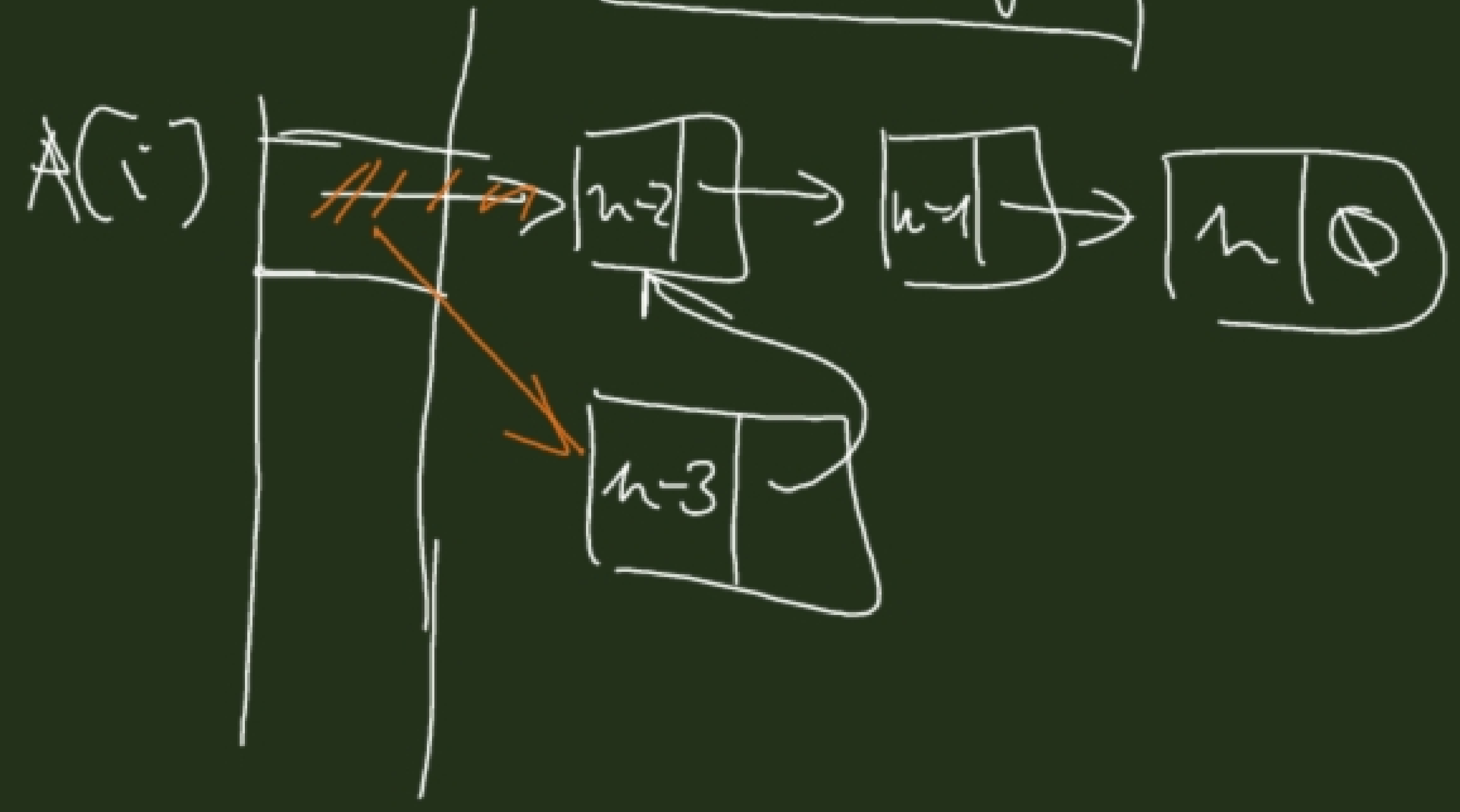
Exercise: algorithm to create the adjacency list representation of K_n.





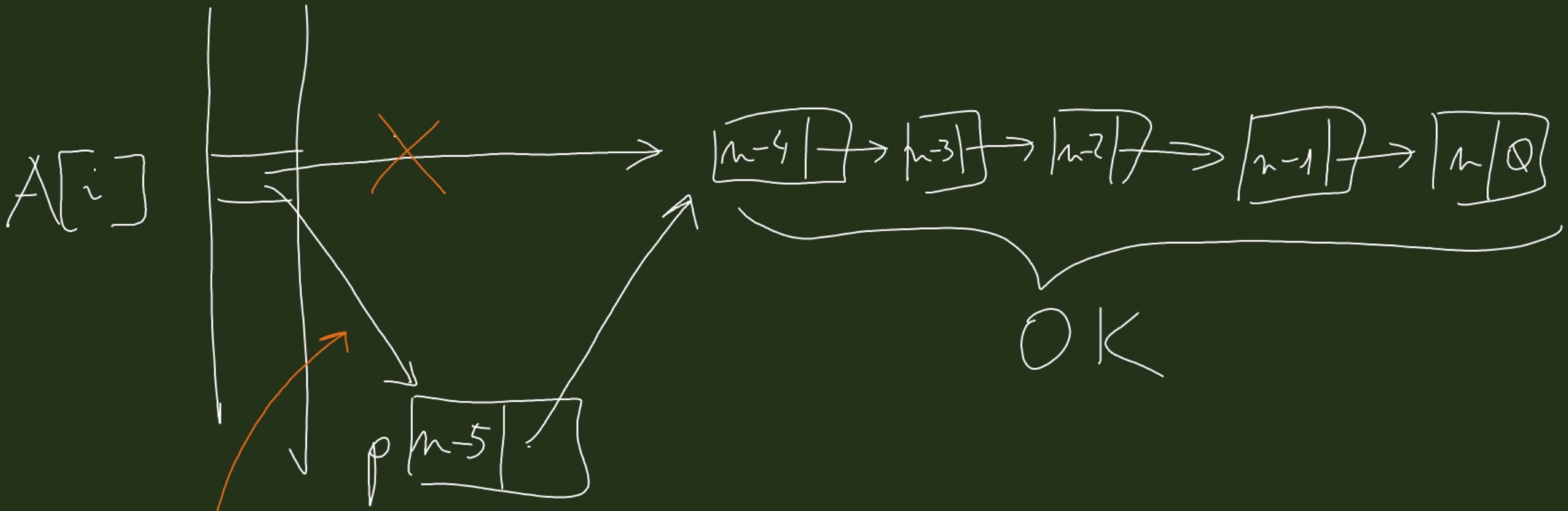
Edge

Edge
+ v : ℕ
+ next : Edge*



$\text{Adj.List}(n : \mathbb{N}) : \text{Edge}^*[n]$

```
A := new Edge*[n]
i = 1 to n
    A[i] := ∅
    j = n downto 1
        i ≠ j
        then:
            p := new Edge
            p→v := j
            p→next := A[i]
            A[i] := p
        else:
            SKIP
return A
```

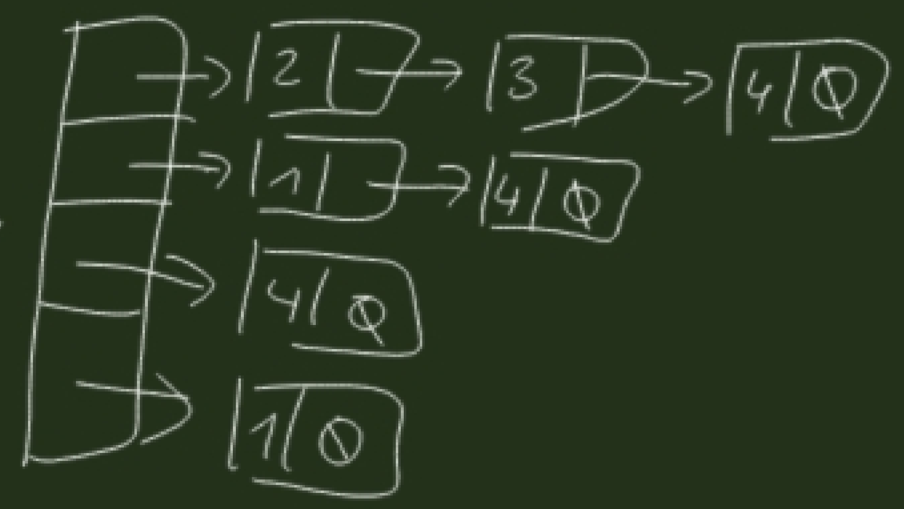


$p \rightarrow next := A[i]$

$A[i] := p$

Design algorithm adj. matrix $\longrightarrow$ adj. list

$$\begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix} \longmapsto$$



AdjMtxToAdjList$(A : \mathbb{N}\ bit[n,n])$: Edge$^*[n]$

Adj := new Edge$^*[n]$

$i = 1$ to n

A[i] := ∅

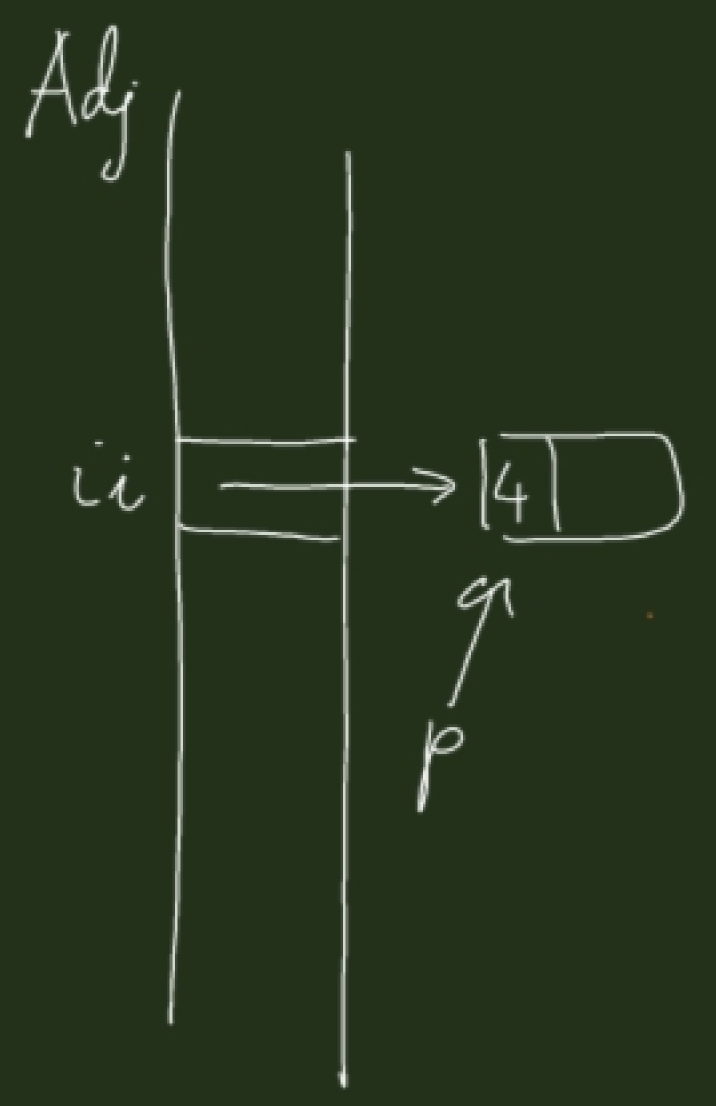
$j = n$ downto 1

A[i,j] = 1

p := new Edge p→v := j p→next := A[i] A[i] := p	SKIP

return Adj

Design algorithm: adj. list → adj. mtx



$k := p \to v$
$A[i,k] := 1$

$\mathbb{N}$

AdjListToAdjMtx(Adj/1 : Edge*[n]) : bit[n,n]

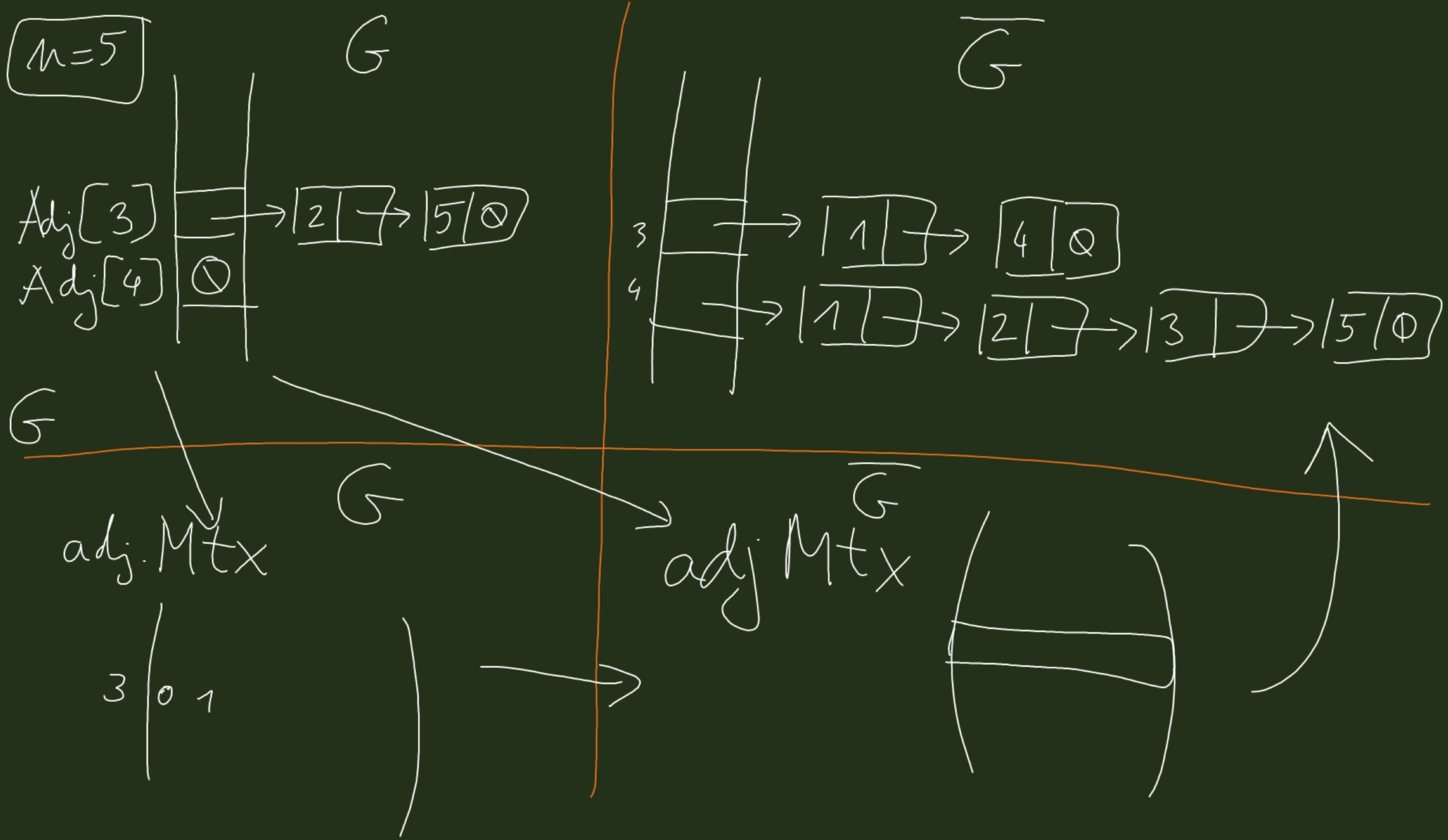
```
A := new bit[n,n]
i = 1 to n
    j = 1 to n
        A[i,j] := 0        } n^2
i = 1 to n
    p := Adj[i]
    p ≠ ∅
        A[i, p->v] := 1
        p := p->next
return A
```

Adj. mtx of the complement of G from Adj. list repr. of G:

$A[i,j] := 1$

$A[i,i] = 0$

$A[i, p \to v] := 0$



n=5
G
Adj[3]
Adj[4]
2
5
$\overline{G}$
3
4
1
4
1
2
3
5
G
G
adj.Mtx
$\overline{G}$
adj Mtx
3
0 1